

Introduction to Macroeconomics (S102016CR)

Spring Semester

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graph TD; A[Fourth Part: Macroeconomic Policies] --> B[AD and AS model (Chapter 10)]; A --> C[Government Policies (Chapter 11)]; A --> D[Inflation and Unemployment (Chapter 12)];
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Fourth Part:
Macroeconomic
Policies

AD and AS
model
(Chapter 10)

Government
Policies
(Chapter 11)

**Inflation and
Unemployment
(Chapter 12)**

Week 12:
Trade-off between
Inflation and Unemployment

- In the session on Money growth and inflation, there was no trade-off between inflation and unemployment.
- We were indeed in **long-run economics**, and the level of economic activity does not depend on prices (classic dichotomy between nominal and real variables)
- In the **short run**, however, prices and real output are linked in the AD-AS model.
- In this chapter, we will explore the short-run trade-off that there exists between inflation and unemployment in the short-run, as well as the role played by inflation expectations in the transition to the long run.
- We end up with some thoughts on the importance of a credible Central Bank to reduce the costs associated with high unemployment levels when the Central Bank implements contractionary monetary policies to fight inflation.

Chapter Contents

- a) The Phillips curve in the long and short-run
- b) The role of expectations and the short-run Phillips curve
- c) Shocks to aggregate supply and the short-run Phillips curve
- d) Credibility of the Central Bank
- e) Summary

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a. The Phillips curve in the long and short-run

In the long run:

- The natural rate of unemployment depends on various labor market features: minimum wage laws, the market power of unions, the role of efficiency wages, and the effectiveness of job search.
- The inflation rate depends on the growth in the quantity of money controlled by the central bank.

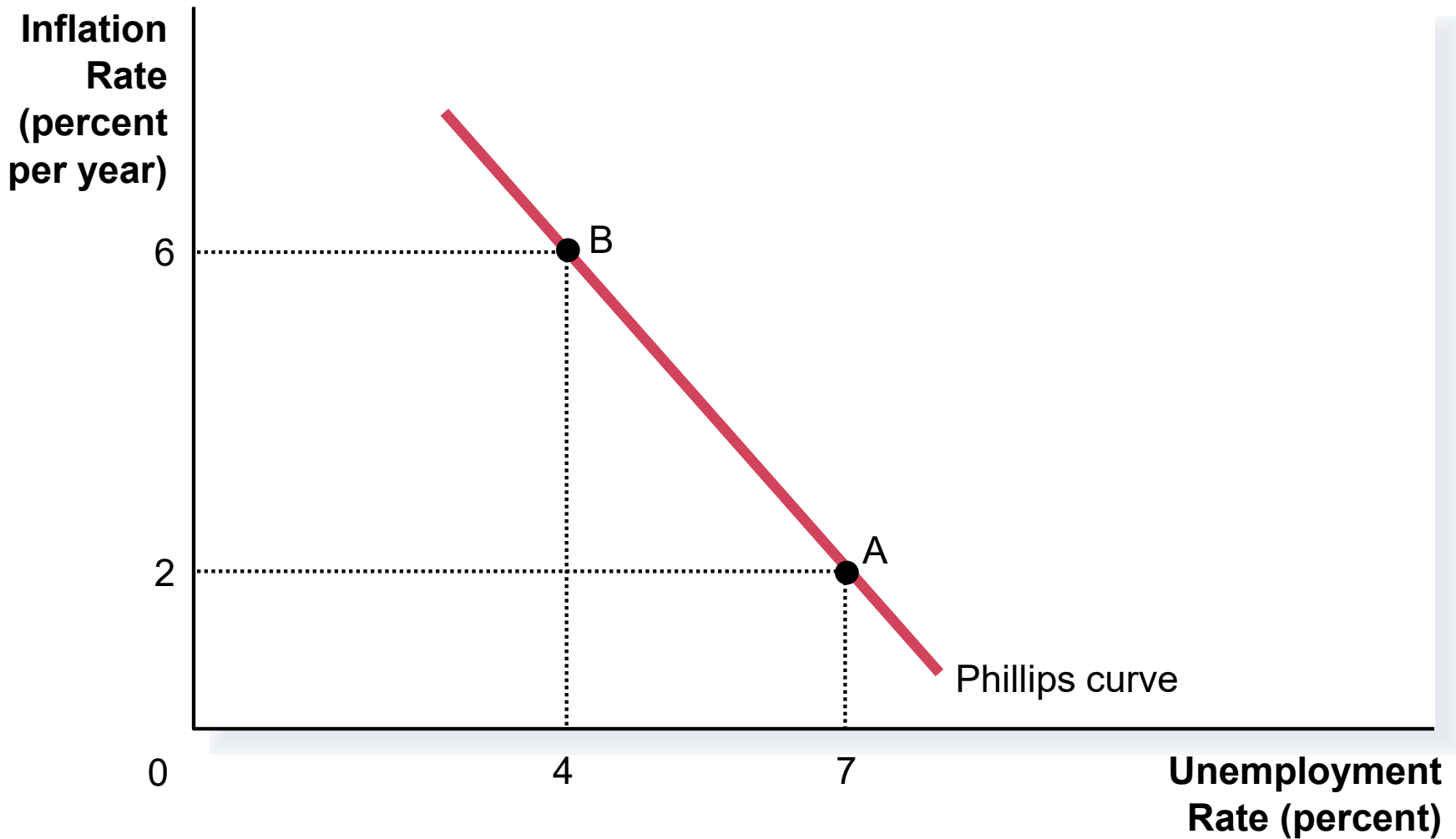
➔ There is no link or trade-off between inflation and unemployment

In the short run, the model of AD-AS tells us that:

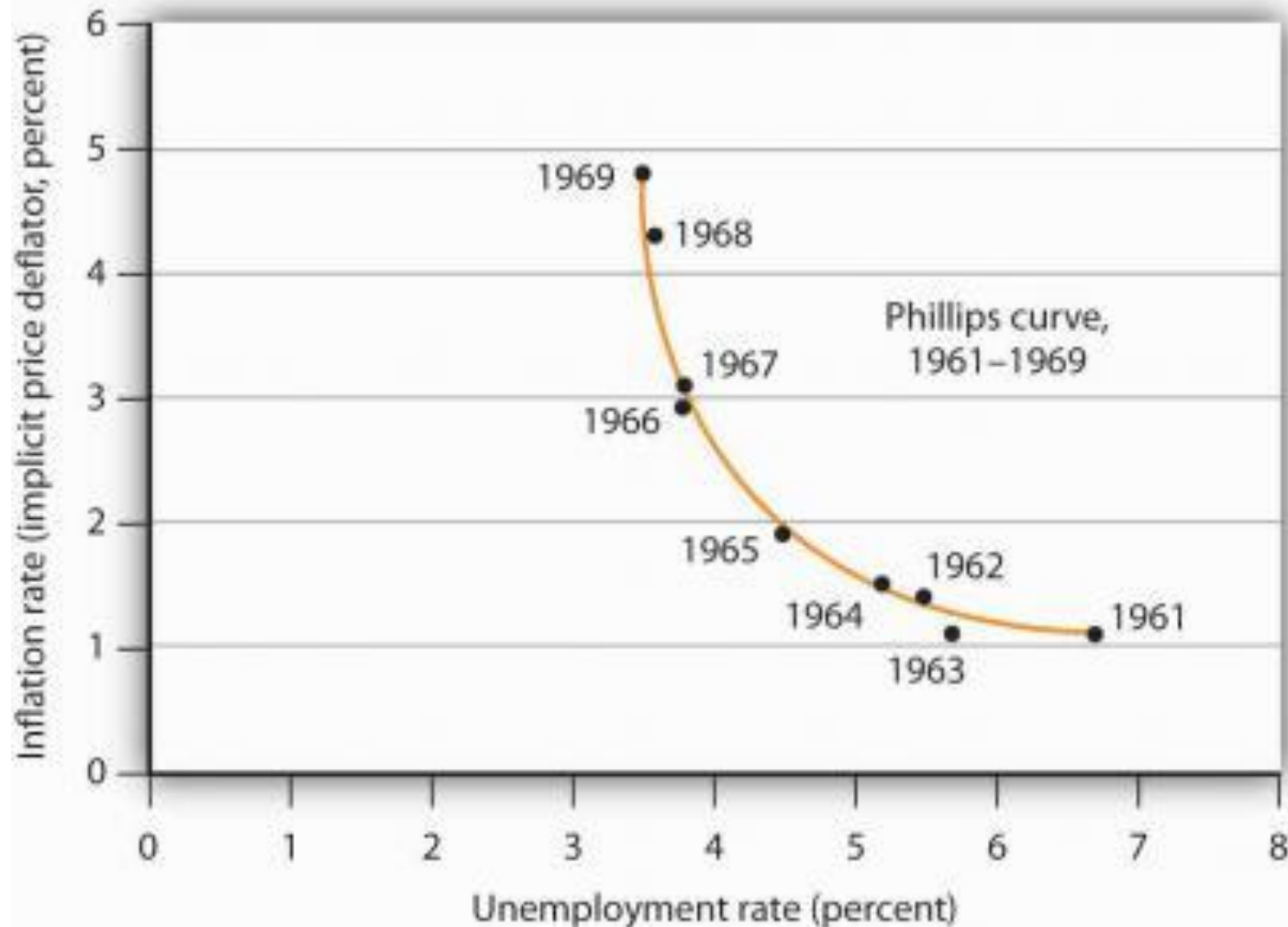
- If policymakers expand aggregate demand, they can lower unemployment, but only at the cost of higher inflation.
- If they contract aggregate demand, they can lower inflation, but at the cost of temporarily higher unemployment.

➔ There is a trade-off between inflation and unemployment.

The *Phillips curve* illustrates the short-run relationship between inflation and unemployment.

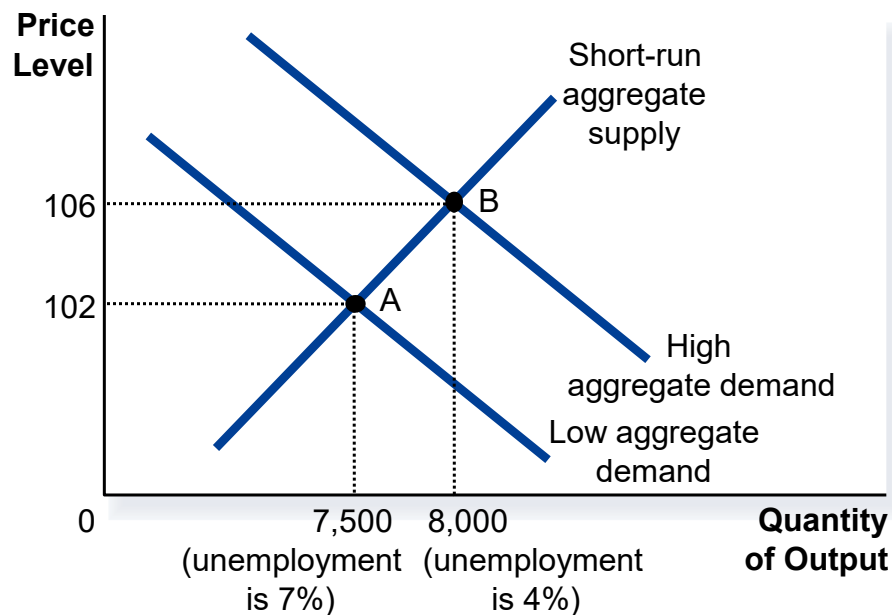


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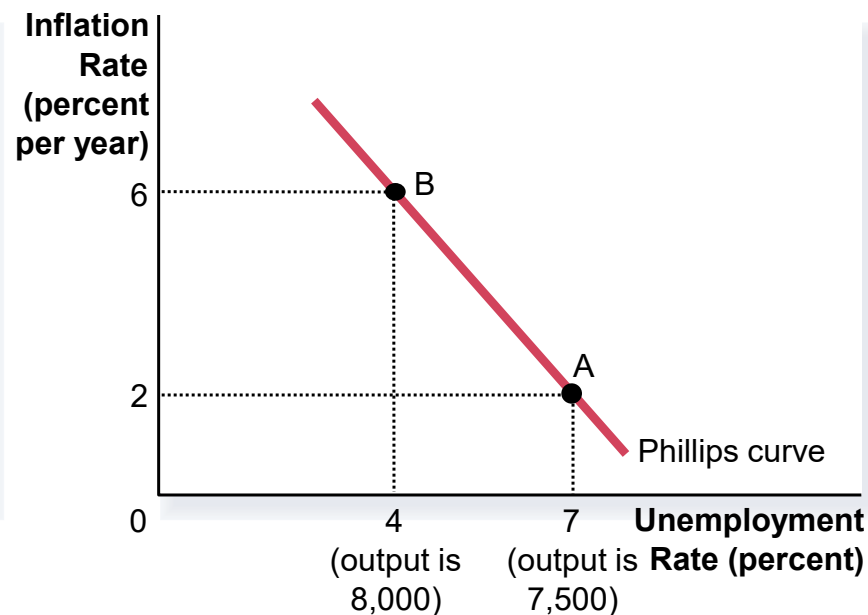


The Keynesian derivation of the Phillips curve using AD and AS

(a) The Model of Aggregate Demand and Aggregate Supply



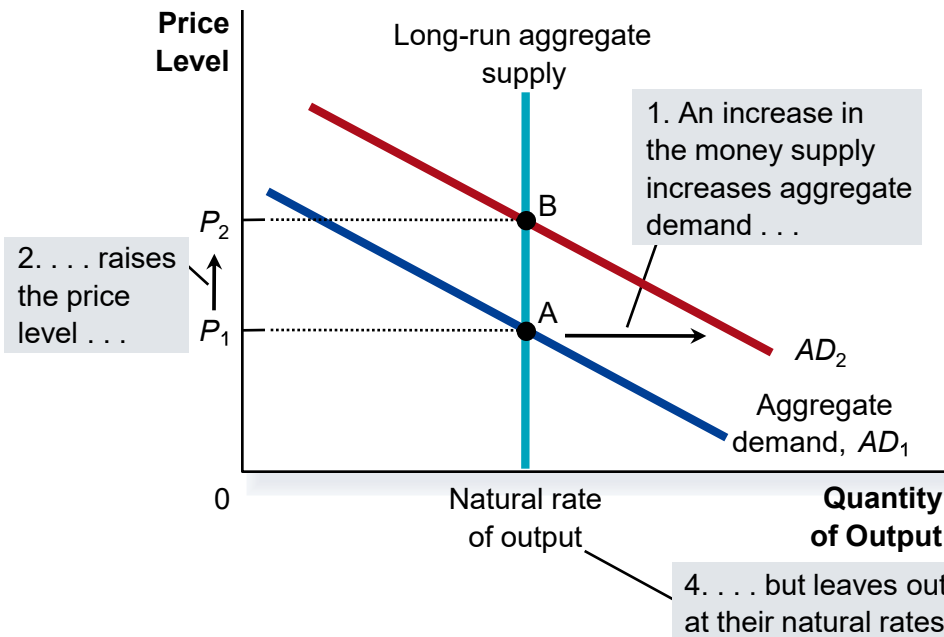
(b) The Phillips Curve



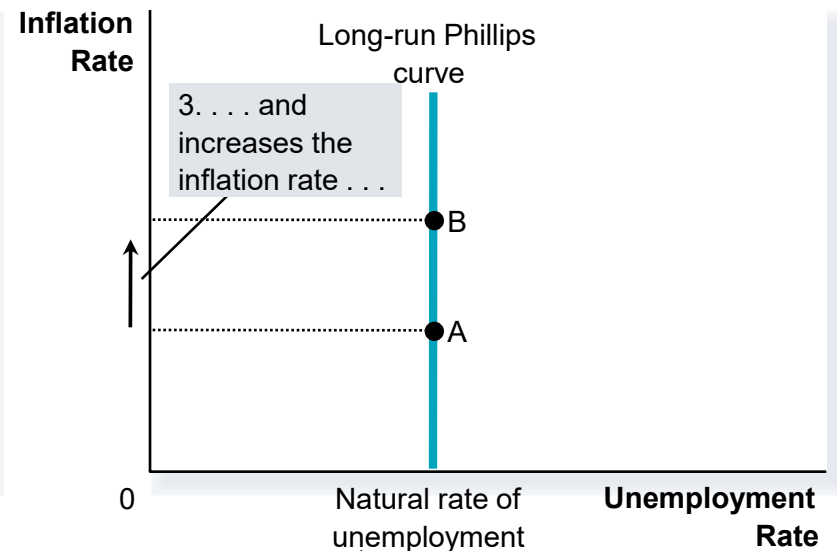
- A positive shock to AD leads to higher prices and higher output in the short run as shown in the AD-AS diagram
- Higher output means less unemployment.
- This yields the negative relationship between inflation and unemployment shown by the Phillips curve.

- In the 1960s, Friedman and Phelps concluded that inflation and unemployment are unrelated in the long run.
 - As a result, the long-run Phillips curve is vertical at the *natural rate of unemployment*.

(a) The Model of Aggregate Demand and Aggregate Supply



(b) The Phillips Curve



Remarks on inflation

Milton Friedman (1966) "Inflation is always and everywhere a monetary phenomenon in the sense that it is and can be produced only by a more rapid increase in the quantity of money than in output".

In the short run, we can think about other sources of inflation:

- **'Cost-Push Inflation'** is a phenomenon in which the general price levels rise (inflation) due to increases in wages and the cost of raw materials.
- **'Demand-Pull Inflation'** is a term used in Keynesian economics to describe the scenario when price levels rise because of an imbalance in the aggregate supply and demand. When the aggregate demand in an economy strongly outweighs the aggregate supply, prices increase.
- **'Imported Inflation'** is due to an increase in the price of imports. As the price of imports increase, prices of domestic goods using imports as raw materials also increases, causing an increase in the general prices of all goods and services. Imported inflation may be caused by foreign price increases or a country's currency depreciation.

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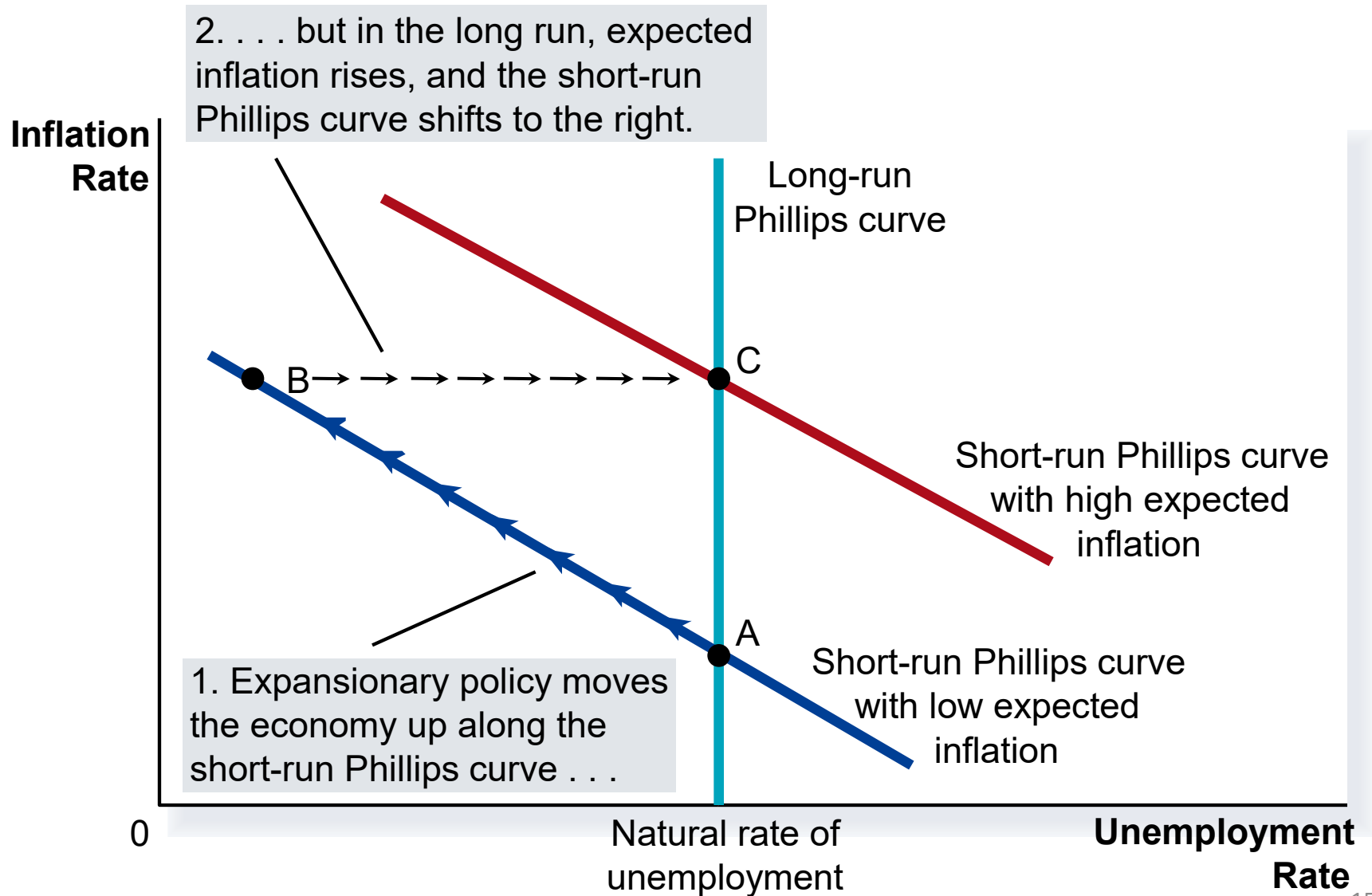
b. The role of expectations and the short-run Phillips curve

- Expected inflation measures how much people expect the overall price level to change.
- In the long run, expected inflation adjusts to changes in actual inflation.
- The central bank's ability to create unexpected inflation exists only in the short run.
 - Once people anticipate inflation, the only way to get unemployment below the natural rate is for actual inflation to be above the anticipated rate.

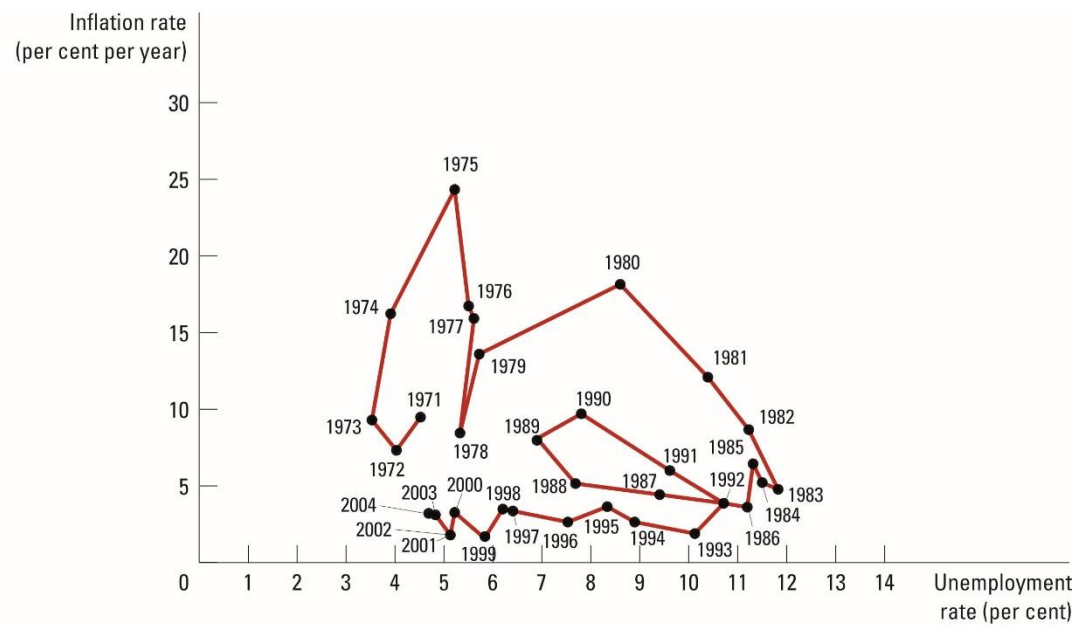
$$\text{Unemployment Rate} = \text{Natural rate of unemployment} - a(\text{Actual inflation} - \text{Expected Inflation})$$

- “ $-1/a$ ” is the slope of the short-run Phillips curve.
- And changes in expected inflation will shift the Phillips curve
- And the increase in expected inflation shifts the Phillips curve to the right, leading to higher levels of unemployment for a given inflation rate.

Expansionary monetary policy: shift from the short-run Phillips curve to the long-run Phillips curve



- The view that unemployment eventually returns to its natural rate, regardless of the rate of inflation, is called the *natural-rate hypothesis*.
- Historical observations support the natural-rate hypothesis.
- Historical events have shown that the short-run Phillips curve can shift due to changes in expectations.
- The stable Phillips curve relationship between inflation and unemployment broke in the early '70s.



Source: UK Office for National Statistics.

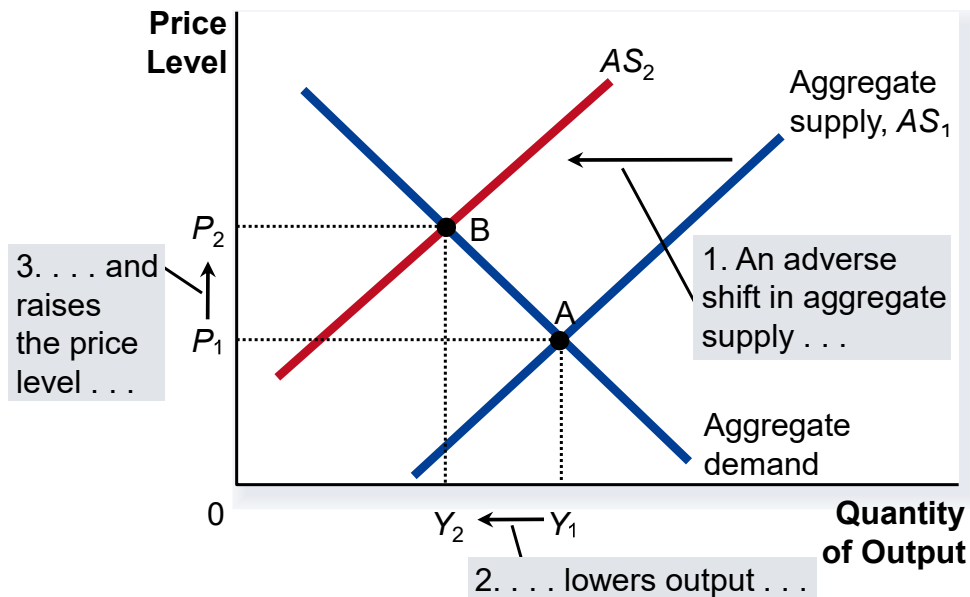
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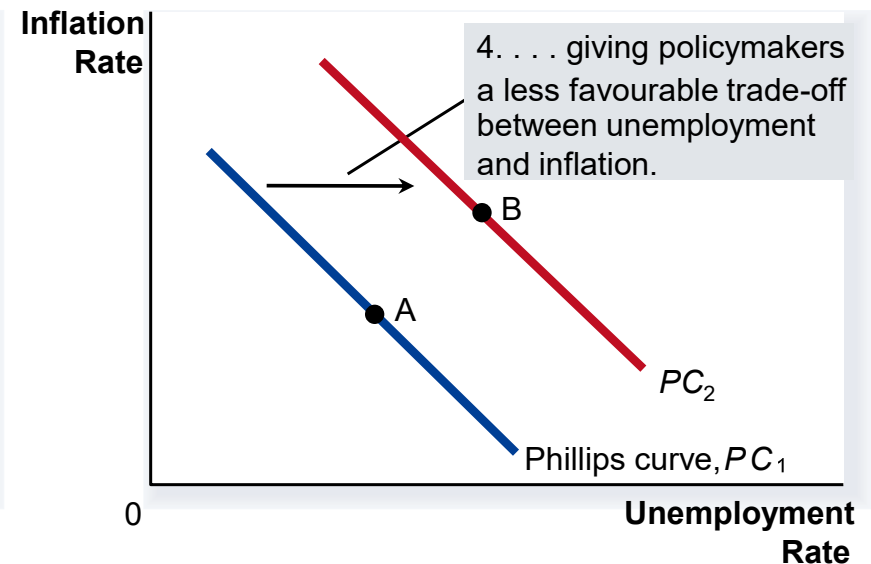
c. Shocks to aggregate supply and the Short run Phillips curve

- The short-run Phillips curve also shifts because of shocks to aggregate supply.
 - Major adverse changes in aggregate supply can worsen the short-run trade-off between unemployment and inflation.
 - An adverse *supply shock* gives policymakers a less favorable trade-off between inflation and unemployment.

(a) The Model of Aggregate Demand and Aggregate Supply



(b) The Phillips Curve



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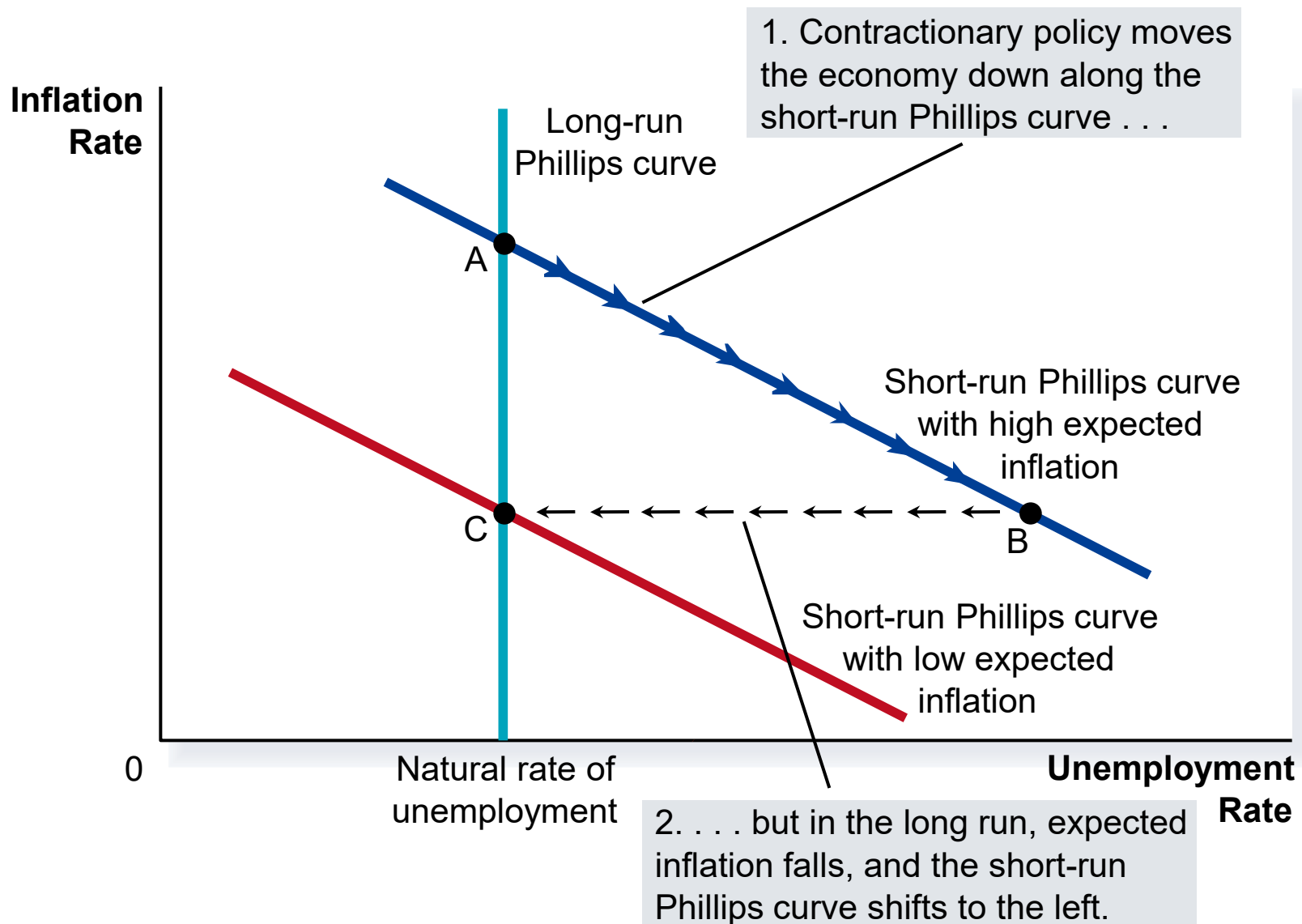
d. Credibility of the Central Bank

- The goal of monetary policy is to control inflation and its costs.
- For this, the Central Bank has different potential tools: money supply and the exchange rate can be considered intermediate monetary policy targets, as they will eventually affect prices.
- But neither the money supply nor the exchange rate is under the direct control of the Central Bank.
- The implication is that the government should target the inflation rate directly and use interest rates to achieve the target.

The economic importance of a credible monetary policy

- When governments try to cut inflation by pursuing a contractionary monetary policy (raising interest rates), it must be done in a credible way (which highlights the importance of the independence of the Central Bank)
 - Why? The more credible the contractionary policy, the quicker expectations will adjust to lower inflation, shifting the short-run Phillips curve to the left and reducing the cost of unemployment induced by contractionary policies.

- The more credible the contractionary policy, the fastest we move from B to C, reducing the costs associated with high unemployment.



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e. Summary

- The Phillips curve describes a negative relationship between inflation and unemployment.
- By expanding aggregate demand, policymakers can choose a point on the Phillips curve with higher inflation and lower unemployment.
- By contracting aggregate demand, policymakers can choose a point on the Phillips curve with lower inflation and higher unemployment.
- The trade-off between inflation and unemployment described by the Phillips curve holds only in the short run.
- The long-run Phillips curve is vertical at the natural rate of unemployment.
- The short-run Phillips curve also shifts because of shocks to aggregate supply.
- An adverse supply shock gives policymakers a less favorable trade-off between inflation and unemployment.
- When the central bank contracts growth in the money supply to reduce inflation, it moves the economy along the short-run Phillips curve.
- This results in temporary high unemployment.

- The cost of disinflation depends on how quickly expectations of inflation fall.
- Thus, a credible CB is crucial to reduce the costs in terms of unemployment associated with contractionary monetary policies
- Inflation targeting is used by most Central Banks today. And the tool used is interest rates which the CB lends to commercial banks.
- This is because the CB has direct control over these interest rates, whereas it does not directly control the money supply, exchange rates, or the interest rates commercial banks lend to the public.